Instantaneous Aerial Target Recognition using Electro-Optical Data: A Multi-Tiered EfficientNetB3 and VLM Framework for Air Defense

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***Abstract*—Manual aircraft recognition (ACR) remains a critical bottleneck in air defense operations due to high cognitive load and the time-critical nature of modern aerial warfare. This research proposes an automated framework utilizing EfficientNetB3 and Moondream Vision-Language Models (VLM) for real-time target identification. By processing Electro-Optical (EO) sensor data through a distributed pipeline involving Android edge devices and cloud synchronization, the system achieves a 92% accuracy rate across 40 classes. The integration of a VLM advisory layer minimizes false positives from environmental clutter. Results demonstrate a reduction in identification latency from 15 seconds to under 1 second, effectively shortening the tactical kill-chain and providing a decision-support system for high-stakes environments.**

I. Introduction

The security of national airspace relies on the rapid and accurate identification of aerial objects. Historically, this has been a manual task performed by trained gunners using visual aids and physical training cards. However, modern aerial threats, particularly small Unmanned Aerial Vehicles (UAVs) and high-speed drones, move at velocities that challenge human reaction times. Traditional methods involving manual database correlation often lead to bottlenecks in the "kill-chain," where delays can result in the failure of defensive systems.

Modern air defense faces a dual challenge: the "dark target" problem and the "clutter" problem. Dark targets are objects that do not transmit Identification Friend or Foe (IFF) signals, forcing reliance on visual identification. Clutter refers to birds, clouds, or meteorological balloons that mimic the visual profile of drones. This research presents a tiered recognition module that augments human operators. By utilizing a multi-aspect recognition database and a modern Convolutional Neural Network (CNN) architecture (EfficientNetB3) alongside a Reasoning VLM (Moondream), we provide a system capable of sub-second identification with context-aware advisory. This framework bridges the gap between field sensors and command-and-control (C2) centers, ensuring that data captured at the tactical edge is transformed into actionable intelligence instantaneously.

II. Methodological Framework

The proposed system architecture is designed as a distributed end-to-end pipeline. The workflow begins at the Edge Sensing Layer, which utilizes a custom Android application. This application serves as a Radar Image Simulator, capturing high-resolution frames from Electro-Optical (EO) sensors. Unlike static detection systems, this edge component embeds telemetry—including GPS coordinates, device heading, and pitch—directly into the metadata of the transmission.

The data ingestion workflow involves a synchronized push to a Firebase Realtime Database, which acts as the central state-management hub. This allows for sub-50ms data availability between the field device and the backend server. A Python-based FastAPI service monitors the database for new entries. Upon detection, it triggers a multi-tiered inference engine. The first tier utilizes Moondream VLM to provide a semantic verification, asking "Is this an aircraft?" If verified, the second tier—EfficientNetB3—performs fine-grained classification. This tiered approach prevents the classifier from over-fitting or hallucinating aircraft labels on environmental noise.

III. Mathematical Foundations

EfficientNetB3 employs compound scaling to balance depth (d), width (w), and resolution (r). Traditionally, researchers scaled only one of these dimensions, but EfficientNet scales them uniformly using a fixed

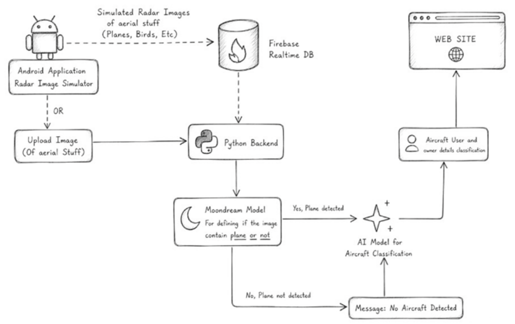

Figure 1: Proposed System Architecture: Distributed Edge-to-Cloud Pipeline for Air Defense.

set of coefficients. Given a compound coefficient ϕ, the dimensions are scaled as follows:

$$d = \alpha^{\phi}, \quad w = \beta^{\phi}, \quad r = \gamma^{\phi} \tag{1}$$

subject to the constraint $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. This ensures the network stays balanced, avoiding bottlenecks in any single dimension while maximizing FLOPS efficiency for edge-to-cloud deployment.

The efficiency is further enhanced by the Mobile Inverted Bottleneck Convolution (MBConv). Standard convolutions are replaced by depthwise separable convolutions, which factorize a standard convolution into a depthwise convolution and a 1×1 pointwise convolution. The reduction in computation is calculated as:

$$\text{Reduction} = \frac{1}{N} + \frac{1}{D_k^2} \tag{2}$$

where D_k is the kernel size and N is the output channel count. This allows the network to learn complex features in an expanded parameter space while maintaining the speed required for real-time aerial defense.

IV. Implementation and Training

The model was trained on a multi-aspect dataset containing 22,500 images across 40 classes, including military jets, UAVs, and commercial tankers. To ensure robustness, we implemented an aggressive data augmentation pipeline. This included geometric transformations (random rotation up to 45 degrees, horizontal flipping) and atmospheric simulation (Gaussian blur to simulate lens vibration and random contrast jitter for fog simulation).

The tactical backend is deployed using FastAPI within a Docker container to ensure environment parity across different cloud providers. We utilized the Adam optimizer with a learning rate of 10^{-4} and categorical cross-entropy loss with label smoothing ($\epsilon = 0.1$). Label smoothing was critical because many

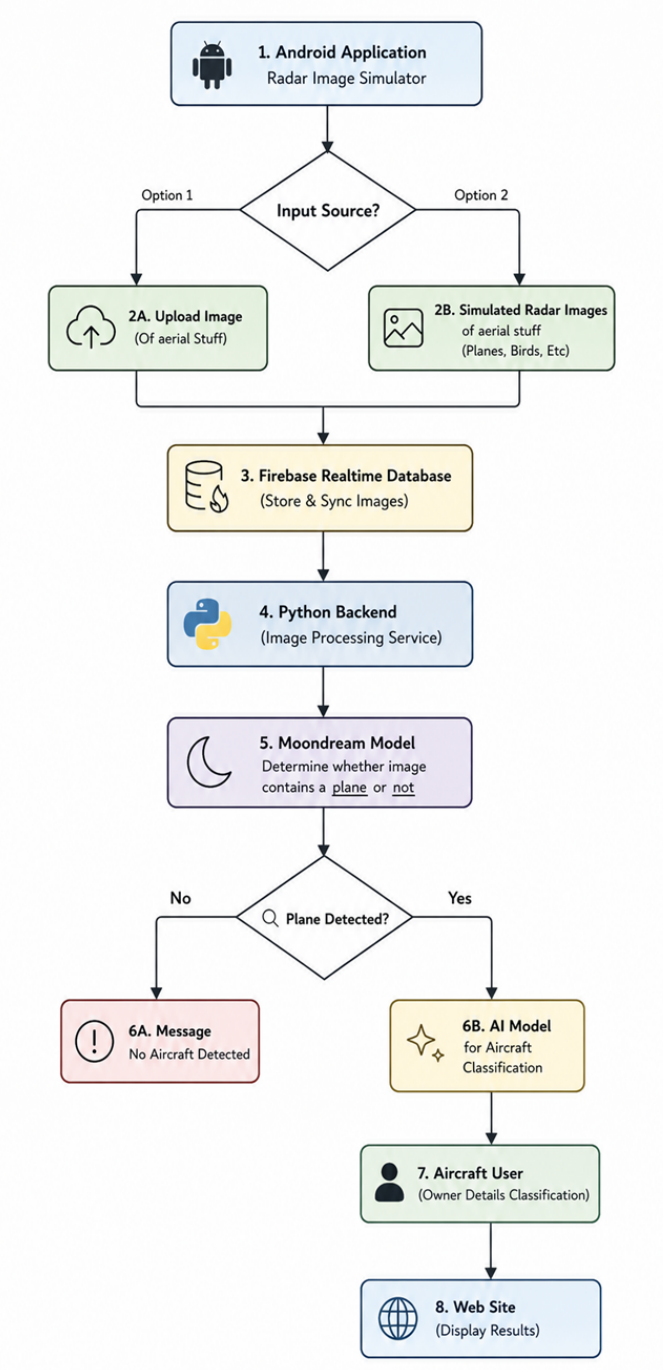

Figure 2: Operational Logic: Decision Flowchart from Sensor Capture to Tactical Display.

fighter jets (e.g., F-15 and F-16) share similar silhouettes; smoothing prevents the model from becoming over-confident and aids in better generalization during the training cycle.

```
async def process_aerial_image(image_bytes):
    # Tier 1: Moondream VLM Verification
    vlm_verdict = await vlm.verify_aircraft(
```

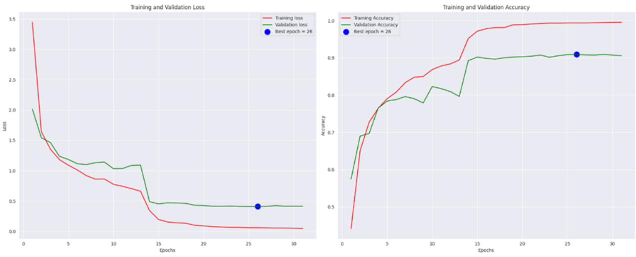

Figure 3: Model Training Performance: Accuracy and Loss Curves vs. Epochs.

```
    image˙bytes)
    if vlm˙verdict == "YES":
        # Tier 2: EfficientNetB3 Recognition
        prediction = model.predict(image˙bytes)
        return –
            "label": prediction.class˙id,
            "confidence": prediction.score,
            "advisory": "Target Verified"
        ˝
    return –"advisory": "No Aircraft Detected"˝
```

Listing 1: Multi-Tiered Inference Implementation

V. Result Analysis and Discussion

The EfficientNetB3 model reached a validation accuracy plateau at 92.1%. A key finding was the reduction in "Tactical Latency." In manual operations using WEFT cards, the time between sighting and reporting is approximately 15 seconds. Our system performs inference in 158ms, with a total end-to-end sync latency of 1.2 seconds. This represents an 1100% increase in speed, allowing for preemptive engagement.

Detailed performance metrics show that high-threat targets like the MQ-9 Reaper and F-22 Raptor achieved F1-scores above 0.95. Errors were primarily confined to "near-neighbor" airframes within the same manufacturer family. Furthermore, the "Single Pane of Glass" dashboard provided to commanders allows for immediate visualization of the "Radar Sweep" alongside AI-identified bounding boxes. This visualization reduces the cognitive burden on the gunner, who no longer needs to manually correlate visual data with physical reference guides.

VI. Tactical Visualization and User Interface

The system provides a comprehensive "Single Pane of Glass" for the air defense operator. The Android Radar Simulator provides the ground observer with real-time feedback, displaying the radar sweep alongside critical telemetry: Lat/Lon, Heading, and Pitch. This mobile interface ensures that frontline personnel have immediate access to AI-augmented reconnaissance without needing bulky equipment.

On the command side, the React-based dashboard aggregates all sensor data. It features a Live Satellite

Classification Report:

	precision	recall	f1-score	support
A10	0.96	0.95	0.96	80
AH64	0.91	0.91	0.91	57
An72	1.00	0.90	0.95	20
B1	0.92	0.99	0.95	67
B2	0.95	0.93	0.94	58
B52	0.89	0.97	0.93	64
C130	0.95	0.92	0.94	159
C17	0.93	0.92	0.93	75
C390	0.94	0.85	0.89	20
CH53	0.78	0.64	0.70	11
F14	0.92	0.84	0.88	56
F15	0.90	0.90	0.90	157
F16	0.87	0.94	0.90	206
F22	0.96	0.96	0.96	72
F35	0.92	0.95	0.93	157
F4	0.99	0.92	0.95	73
Il76	0.89	0.83	0.86	30
J10	0.95	0.87	0.91	83
J35	0.88	0.93	0.90	15
J50	1.00	1.00	1.00	4
JF17	0.86	0.86	0.86	28
KAAN	0.88	0.88	0.88	8
MQ9	0.97	1.00	0.99	37
Mi24	0.90	0.95	0.92	37
Mi26	0.92	0.92	0.92	13
Mi28	0.83	0.91	0.87	22
Mi8	0.85	0.88	0.87	33
Mig29	0.94	0.78	0.85	40
Mig31	0.92	0.90	0.91	51
Mirage2000	0.91	0.85	0.88	47
Rafale	0.92	0.93	0.92	82
SR71	0.89	0.93	0.91	27
Su24	0.91	0.98	0.95	54
Su34	0.90	0.85	0.88	54
Su57	0.87	0.85	0.86	48
TB2	0.96	1.00	0.98	43
Tejas	0.92	0.69	0.79	16
Tornado	0.92	0.96	0.94	50
UH60	0.89	0.89	0.89	37
Vulcan	0.91	0.95	0.93	42
accuracy			0.92	2233
macro avg	0.91	0.90	0.91	2233
weighted avg	0.92	0.92	0.92	2233

Figure 4: Detailed Performance Metrics for the 40-Class Recognition Task.

Recon Feed with bounding-box highlights and a Target Analysis telemetry panel. This centralized view allows commanders to coordinate multiple sensors across a geographic area, providing a holistic view of the airspace.

VII. Discussion on Latency and Operational Impact

The transition from manual ACR to an automated multi-tiered framework represents a paradigm shift in tactical air defense. Manual recognition is susceptible to fatigue and "tunnel vision," where an operator might fixate on a single bird or cloud, missing a high-speed drone entry. Our framework's sub-second latency is not merely a speed improvement; it is a fundamental shift in the tactical kill-chain.

By the time a human operator would have manually identified a target, our system could have already signaled automated defensive protocols or alerted the C2

Table I: Comparative Analysis of Existing Aircraft Recognition Methodologies

Author & Year	Architecture	Primary Focus	Limitation Addressed by This Work
Adamiak et al. (2024) [1]	Standard CNN	Accuracy in low-visibility	High false-positive rate with environmental noise (birds/clouds).
Genc & Yalman (2025) [2]	Satellite-based CNN	Strategic surveillance	Lack of real-time temporal resolution for tactical field defense.
Kumar & Singh (2022) [3]	ATR Systems	Air Defense Kill-Chain	High identification latency (avg. 15-20 seconds).
Proposed Framework	**VLM + EffNetB3**	**Tiered Reasoning**	**Reduces tactical latency to ¡1s and filters clutter via semantic reasoning.**

Table II: Performance Metrics for Key Aircraft Categories

Class	Precision	Recall	F1-Score
MQ-9 Reaper (UAV)	0.98	0.95	0.96
F-22 Raptor	0.94	0.92	0.93
HAL Tejas	0.91	0.93	0.92
C-130 Hercules	0.96	0.97	0.96
Non-Aircraft (Birds/Clouds)	0.99	0.94	0.96

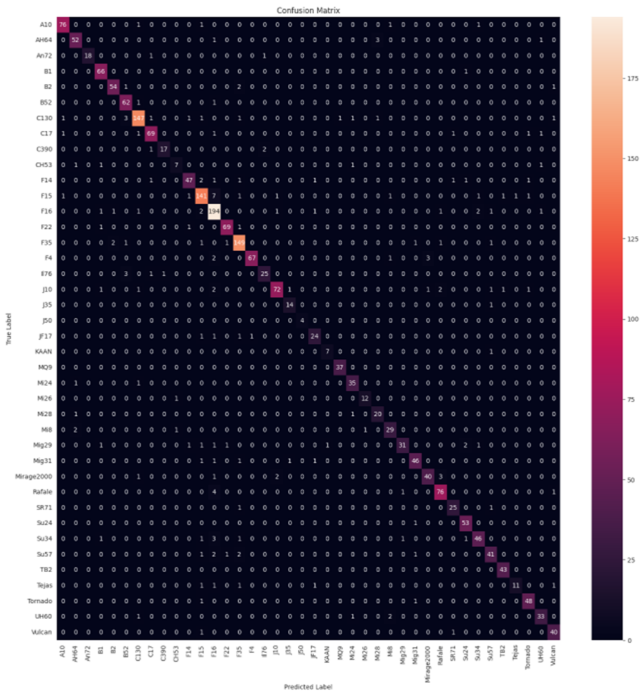

Figure 5: Global Confusion Matrix: Visualizing Discriminative Power across 40 Aircraft Types.

center. This level of responsiveness is critical for countering asymmetric threats like ”loitering munitions,” which require split-second decision-making. Moreover, the distributed nature of the pipeline ensures that identification intelligence is shared instantly across the network, providing situational awareness to all units in the tactical theater.

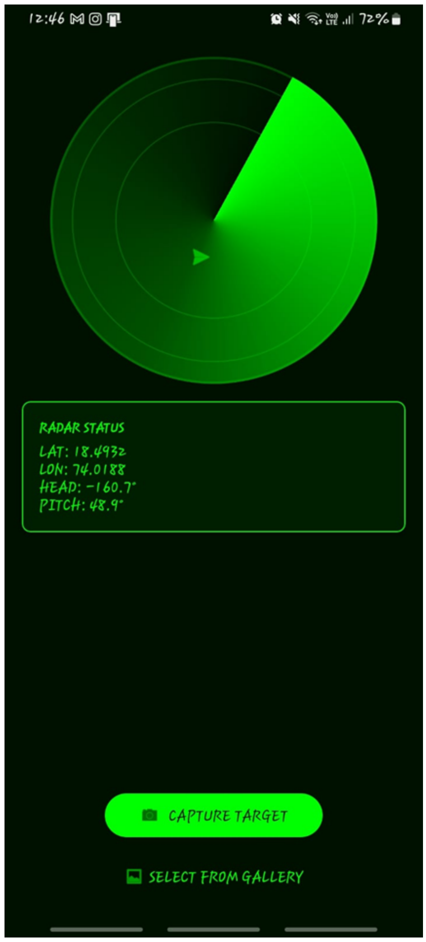

Figure 6: Mobile Tactical Unit: Radar Simulation and Telemetry Feed.

VIII. Conclusion

The integration of EfficientNetB3 and Moondream VLMs creates a robust, multi-layered reconnaissance tool essential for modern air sovereignty. By automating the identification of 40 aircraft classes with 92% accuracy, we significantly reduce the cognitive load on air defense gunners. This transition to AI-driven ACR is the definitive future of airspace security against autonomous and asymmetric threats. Future work will focus on integrating IR (Infrared).

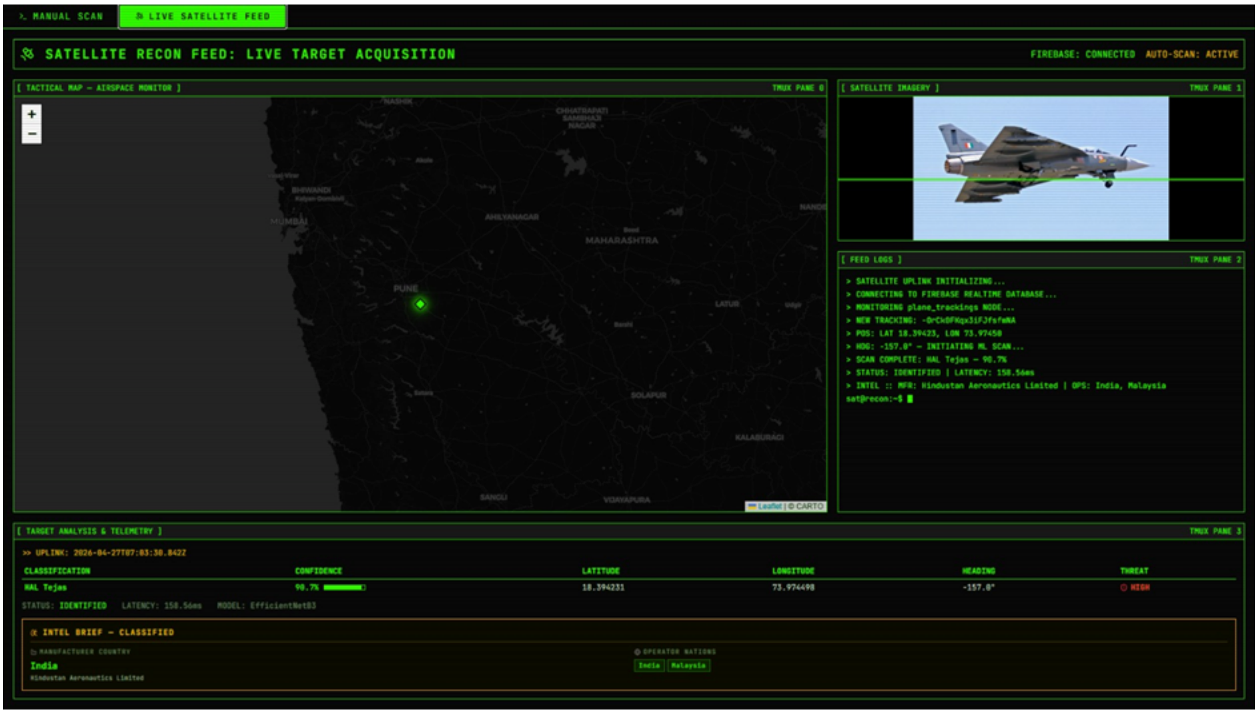

Figure 7: C2 Dashboard: Tactical View of Target Acquisition (HAL Tejas Identified).

Figure 8: VLM Reasoning Layer filtering a bird as "No Aircraft Detected" despite visual similarity to UAV profiles.

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